

STUDYSPHERE

Class 9 Science

CHAPTER 1 — CELL

The Fundamental Unit of Life

Detailed Revision Notes • Concepts • Comparisons • Activities • Exam Practice

These are original study notes written in simple language for revision. They cover the major cell concepts, important organelles, transport, cell division and competency-based practice.

How to use: First understand the concept, then revise the comparison tables, practise diagrams, and finally attempt the questions without looking at the answers.

1. Cell: The Basic Unit of Life

A **cell** is the basic structural and functional unit of a living organism. Structural means that cells form the body of organisms. Functional means that important life processes occur within cells.

Unicellular and multicellular organisms

- **Unicellular:** made of one cell; the same cell performs all essential life functions. Examples include Amoeba and many bacteria.
- **Multicellular:** made of many cells; different cells can become specialised for different jobs. Humans and plants are examples.

Division of labour

In multicellular organisms, cells can specialise. Groups of similar cells form tissues, tissues form organs, and organs cooperate in organ systems. This specialisation makes complex organisms more efficient.

Basic parts of a typical eukaryotic cell

- Plasma membrane — regulates exchange with the surroundings.
- Cytoplasm — medium containing many cellular structures and reactions.
- Nucleus — contains genetic material and helps regulate cell activity.
- Organelles — specialised structures with particular functions.

Exam focus: Never write that every cell has exactly the same structures. Cell types differ in structure and function.

2. Prokaryotic and Eukaryotic Cells

Cells can be broadly organised as prokaryotic or eukaryotic. The most important distinction at this level is the presence or absence of a membrane-bound nucleus.

Feature	Prokaryotic	Eukaryotic
Nucleus	No membrane-bound nucleus	Membrane-bound nucleus
Size	Generally smaller	Generally larger
Membrane-bound organelles	Absent	Present
Examples	Bacteria	Plant and animal cells

Important clarification

Prokaryotic cells do contain genetic material. The key point is that it is not enclosed inside a membrane-bound nucleus. This is a common exam trap.

Why organisation matters

Eukaryotic cells have specialised compartments. Compartmentalisation allows different biochemical processes to occur in suitable locations, improving organisation and control.

3. Plasma Membrane

The **plasma membrane** is the thin boundary surrounding the cell. It separates the cell contents from the external environment and controls the movement of many substances.

Selective permeability

The membrane is selectively permeable: some substances can cross more readily than others. This helps the cell maintain suitable internal conditions.

Diffusion

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration. It can occur without the cell directly supplying energy for that movement.

Osmosis

Osmosis is the movement of water through a selectively permeable membrane from a region of higher water availability to a region of lower water availability.

Effects on cells

- In a dilute external solution, water may enter a cell.
- In a concentrated external solution, water may leave a cell.
- In an isotonic condition, there is no net movement of water, although molecules continue moving in both directions.

Remember: Diffusion can involve different particles; osmosis specifically refers to water movement through a selectively permeable membrane.

4. Cell Wall and Osmosis in Plants

Plant cells have a **cell wall** outside the plasma membrane. It provides support, protection and helps the cell retain its shape.

Cell wall

- It is mainly associated with cellulose in plant cells.
- It gives mechanical strength.
- It protects the cell and helps it resist excessive expansion.
- It is different from the plasma membrane, which controls selective transport.

Plasmolysis

When a plant cell loses water in a sufficiently concentrated external solution, the cell contents can shrink away from the cell wall. This is called **plasmolysis**.

Plant cell vs animal cell

Feature	Plant cell	Animal cell
Cell wall	Present	Absent
Chloroplasts	Present in photosynthetic cells	Absent
Large central vacuole	Usually prominent	Usually smaller
Shape	Often more fixed	Often more variable

Application: Water movement across membranes explains why plant tissues can become firm or limp depending on their water status.

5. Nucleus, DNA, Genes and Chromosomes

The **nucleus** is a membrane-bound organelle that contains genetic material and helps regulate many cell activities.

Main nuclear structures

- Nuclear envelope — surrounds the nucleus.
- Nuclear pores — allow regulated exchange between nucleus and cytoplasm.
- Chromatin — DNA associated with proteins.
- Nucleolus — region involved in production of ribosomal components.

DNA → genes → chromosomes

DNA carries hereditary information. Genes are functional segments of DNA. DNA is organised with proteins as chromatin, and during cell division the chromatin becomes highly condensed into chromosomes.

Why is the nucleus important?

- 1 It stores genetic information.
- 2 It helps regulate cellular activities.
- 3 It plays an important role during cell division and inheritance.

Exam line: Chromosomes are associated with hereditary information because they contain DNA, and DNA contains genes.

6. Mitochondria and Ribosomes

Cells need energy for transport, growth, repair, synthesis and other activities. They also need to make proteins that perform many cellular functions.

Mitochondria

Mitochondria are major sites of aerobic cellular respiration and energy transformation. They have an outer and inner membrane and also contain their own genetic material and ribosomes.

Ribosomes

Ribosomes are structures involved in **protein synthesis**. They may occur free in the cytoplasm or be attached to rough endoplasmic reticulum.

Quick table

Structure	Main function
Nucleus	Genetic information and regulation
Mitochondria	Major site of aerobic respiration
Ribosomes	Protein synthesis

Memory idea

Nucleus = control and genetic information; mitochondria = energy transformation; ribosomes = protein-making machinery.

7. Endoplasmic Reticulum and Golgi Apparatus

The endomembrane system helps cells synthesise, modify, transport and distribute many substances.

Endoplasmic reticulum (ER)

- **Rough ER:** has ribosomes and is strongly associated with protein synthesis and processing.
- **Smooth ER:** lacks attached ribosomes and is associated with lipid synthesis and other metabolic functions.
- The ER also forms an internal membrane network for movement of materials.

Golgi apparatus

The Golgi apparatus receives materials, modifies them, sorts them and packages them into vesicles. It is especially important in processing and distribution of proteins and lipids.

Simplified pathway

Rough ER → transport vesicles → Golgi → modification/sorting → destination

Lysosomes

Lysosomes contain digestive enzymes and help break down certain materials and worn-out cell components. They contribute to cellular cleaning and recycling.

Vacuoles

Vacuoles are membrane-bound storage compartments. A large central vacuole is characteristic of many plant cells and contributes to storage and internal pressure.

8. Plastids and Chloroplasts

Plastids are important organelles in plant cells. Some contain pigments and participate in photosynthesis, while others have storage roles.

Chloroplast

Chloroplasts contain chlorophyll and are the major sites of photosynthesis in plant cells. They have an internal membrane system and contain their own genetic material and ribosomes.

Photosynthesis — concept

Plants use light energy to help convert carbon dioxide and water into carbohydrate-rich products. Oxygen is released as a by-product in oxygenic photosynthesis.

Other plastids

- **Chromoplasts:** associated with pigments that contribute to colour in many plant parts.
- **Leucoplasts:** generally non-pigmented plastids associated mainly with storage.

Plant-cell diagram checklist

- Cell wall
- Plasma membrane
- Cytoplasm
- Nucleus
- Chloroplasts
- Large vacuole
- Mitochondria
- ER and Golgi

Diagram tip: Use straight label lines, avoid crossing them, and write labels clearly.

9. Cell Division and Growth

Cell division is a controlled process through which cells produce new cells. It is essential for growth, repair and reproduction in different organisms.

Why cells divide

- Growth of multicellular organisms
- Replacement of damaged or old cells
- Asexual reproduction in some organisms
- Development and formation of specialised cells

Mitosis — basic idea

Mitosis produces daughter cells that generally retain the chromosome number of the parent cell. It is important in growth and tissue maintenance.

Simplified sequence

- 1 Chromosomes condense and become organised.
- 2 Chromosomes are arranged and separated by the spindle apparatus.
- 3 Chromosome sets move to opposite sides.
- 4 Nuclei reform and the cytoplasm divides.

Cell cycle

Before division, a cell grows and prepares for division. DNA is duplicated so that genetic information can be distributed to the daughter cells.

Mitosis vs meiosis

Point	Mitosis	Meiosis
Divisions	One	Two
Main role	Growth, repair, replacement	Formation of reproductive cells in sexual reproduction
Chromosome number	Usually maintained	Reduced in resulting cells
Variation	Usually less genetic variation between daughter cells	Daughter cells have genetic combinations

10. Important Activities and Observations

Activity: Observing cells

Microscopes allow us to observe structures that are too small for the unaided eye. A temporary slide must be thin enough for light to pass through and should be handled carefully.

Activity: Demonstrating osmosis

Plant tissues placed in solutions of different concentrations can show changes in mass, firmness or appearance because water moves across cell membranes.

How to write an experiment

- 1 Aim — what is being investigated?
- 2 Materials — important apparatus and substances.
- 3 Procedure — steps in order.
- 4 Observation — what changed?
- 5 Conclusion — which biological concept explains the result?

Application question

A plant cell placed in a concentrated solution becomes shrunken. What explains this observation?

Answer: Water moves out of the cell through the selectively permeable plasma membrane. The cell contents shrink away from the cell wall; this is plasmolysis.

Raisin example

A raisin placed in water can swell because water moves into its cells when the surrounding conditions favour water entry. This is an everyday illustration of osmosis.

11. Exam-Oriented Questions

Very short answer

- 1 What is the basic structural and functional unit of life?
- 2 Name the organelle involved in protein synthesis.
- 3 Name a major site of aerobic respiration.
- 4 Define osmosis.
- 5 Name the pigment important for photosynthesis.

Short answer

- 1 Differentiate diffusion and osmosis.
- 2 Why is the plasma membrane selectively permeable?
- 3 Give two differences between plant and animal cells.
- 4 What is the role of the Golgi apparatus?
- 5 Why are lysosomes important?

Long answer

- 1 Explain the structure and functions of the nucleus.
- 2 Describe the roles of ER and Golgi apparatus in cellular transport and processing.
- 3 Compare prokaryotic and eukaryotic cell organisation.
- 4 Explain why osmosis is important for plant cells.

12. Competency-Based Practice

Case 1: An animal cell is placed in very dilute water and swells. Explain the direction of water movement.

Answer hint: Compare water availability outside and inside the cell and apply osmosis.

Case 2: A cell is producing large quantities of proteins that will be transported elsewhere. Which structures are likely to be strongly involved?

Answer: Ribosomes and rough ER are important for protein production, while the Golgi apparatus helps modify, sort and package many proteins.

Case 3: A cell has genetic material but no membrane-bound nucleus. What type of organisation does this suggest?

Answer: Prokaryotic organisation.

Case 4: A muscle cell has a high energy demand. Which organelle would you expect to be especially important?

Answer: Mitochondria, because they are major sites of aerobic respiration and energy transformation.

Rapid revision

Structure	One-line memory
Membrane	Selective boundary
Cell wall	Plant support and protection
Nucleus	DNA and regulation
Mitochondria	Aerobic respiration
Ribosomes	Protein synthesis
ER	Synthesis/transport network
Golgi	Modify, sort, package
Lysosomes	Digestion/recycling
Vacuole	Storage
Chloroplast	Photosynthesis

13. Final One-Page Revision Sheet

CELL = BASIC UNIT OF LIFE

Plasma membrane: selective barrier → controls movement

Cell wall: plant support and protection

Nucleus: DNA → genes → regulation and heredity

Mitochondria: major site of aerobic respiration

Ribosomes: protein synthesis

ER: internal membrane network; synthesis and transport

Golgi: modification, sorting and packaging

Lysosomes: digestion and recycling

Vacuoles: storage; large central vacuole common in plants

Chloroplast: photosynthesis in plant cells

Cell division: growth, repair and reproduction

Last-minute checklist

- I can define cell, organelle, diffusion and osmosis.
- I can compare plant and animal cells.
- I can compare prokaryotic and eukaryotic cells.
- I know the functions of the major organelles.
- I can explain plasmolysis.
- I can answer application-based questions.
- I can draw and label a basic plant cell and animal cell.

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